Evidence-first selective inference for Bengali hallucination detection

A study of routing, abstention, provenance, and domain shift

Research question

When should exact evidence decide, and when should the system abstain to a language-model judge?

We trace the hypotheses, failed experiments, routing policies, regression gates, and Phase 2 shift.

Touhidul Alam Seyam | MD. Abtahee Kabir | Joyeta Barua Moni | Noore Tamanna Orny

BGC Trust University Bangladesh | Chittagong University of Engineering & Technology

[1] Competition task [3,4] Bengali hallucination and idiom studies [7] Experiment ledger

One binary label hid many different reasoning tasks

The same label had to cover context faithfulness, language knowledge, and outside facts.

Context
entailment

Numbers
dates and ranges

MCQ
option parsing

Language
morphology

Idioms
literal vs figurative

Facts
history and law

A single judge prompt treated all six as the same problem.

That worked as a baseline, but it was prompt-sensitive and weak on culturally specific relations.

[1] Competition task definition [3] BenHalluEval task taxonomy [4] Bengali idiom benchmark

The working hypothesis became a selective cascade

Every stage could answer, or abstain. Abstention was a normal outcome, not a failure.

Normalize
digits, options, aliases

Route
context, numeric, lexical, factual

Verify
exact lineage and typed evidence

Abstain?
resolved label or uncertain tail

Judge
Qwen binary decision

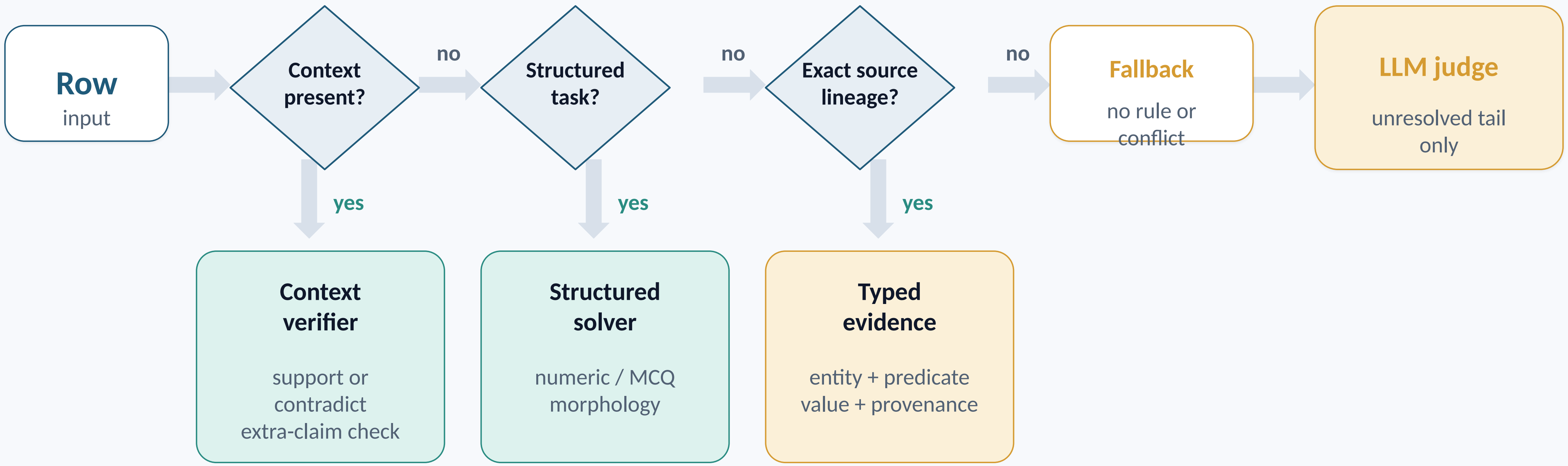
Technical contributions

Selective routing • task solvers • claim verification • typed provenance • offline T4×2

Every deterministic decision records its source family and requested relation.

Routing made abstention an explicit state

A branch could return 0, 1, or abstain. Only abstentions moved to the next branch.



Decision contract branch(row) -> {0, 1, null}; null means abstain, never a guessed label

Structured solvers shared one decision contract

Each solver parsed a task-specific representation, verified the requested relation, then answered or abstained.

Numbers

- normalize Bengali digits
- units / dates / ranges
- **compare requested relation**

MCQ

- extract option set
- map aliases to options
- **validate selected option**

Morphology

- identify requested form
- parse root / suffix / class
- **compare analysis**

Idiom & lexical

- select literal or figurative
- retrieve exact sense
- **compare requested meaning**

Common output {entity, predicate, parsed value, source, verdict} — incomplete or conflicting fields return null

[7] v91 structured solvers, relation schema, and regression cases

A correct fact can still answer the wrong relation

Released sample 4 contains both the film title and year; the prompt asks specifically for the year.

Released labeled example

Context excerpt

‘I ’

Prompt

?

Response I I I I I I I

Released label: 0

Relation check

Entity	Runway	matches
Requested predicate	release_year	expected
Response type	film_title	observed

predicate mismatch → label 0

Similarity finds a related answer. The relation gate rejects it.

Context rows were reduced to claim-level entailment tests

The verifier compared atomic response claims against the relevant context clause, not against global word overlap.

Context
segment into clauses

→

Response
extract atomic claims

→

Anchor
entity, number, negation, relation

→

Alignment
best applicable clause only

→

State
S support / C contradict / U unknown

Decision rule over response claims c1 … cn

Any contradiction	exists i: state(ci) = C	label 0
Complete support	for all i: state(ci) = S; no extra factual claim	label 1
Incomplete alignment	otherwise	abstain

Qwen judged 158 rows, not the whole test set

The fallback was short, deterministic, and biased toward catching hallucinations.

Context row

1 entailment view

Compare the response against the supplied passage.

Any contradiction returns 0.

No-context row

Up to 3 short views

Fact check | exam grading | zero-bias check

Label 1 requires unanimous positive votes.

Qwen3.6-27B Q4_K_P | one-token grammar | fixed seeds and batches

[5] Qwen3.6 model card [6] llama.cpp b9637 [7] Frozen judge configuration

A source record was accepted only after four fields aligned

The trace records why a row was resolved, its source, and the relation that was tested.

Typed evidence record

source_id	stable row or URL
family	law / idiom / QA / fact
entity	normalized subject
predicate	requested relation
value	answer or claim
provenance	dataset + citation

Admission gate

1 Entity
same normalized subject

2 Predicate
same requested relation

3 Value
exact or typed equivalence

4 Conflict
no equal-trust contradiction

accept = entity AND predicate AND value AND no_conflict

Decision trace

route	typed_evidence
source	law_section_42
predicate	commencement date
verdict	1

If any gate fails: abstain (verdict = null)

Versioned comparisons locate the main design pivots

These are cumulative versions, not controlled single-factor ablations.

Version	Public F1	Rows judged	Main methodological change
Baseline	**0.657**	none	character and embedding reference
Uniform Qwen	**0.797**	all rows	general reasoning improved; one policy remained
v6	**0.903**	reduced	source and structured routing
v37	**0.961**	312	stricter provenance gate
v85	**0.966**	193	reproducible two-source anchor
v91	**0.967**	158	typed idiom and predicate guards

The largest observed step coincided with routing: 0.797 → 0.903.

[7] Scored Phase 1 experiment ledger and Kaggle public leaderboard

Decision logs turned failed runs into routing rules

Each negative result changed what later versions were allowed to decide.

Experiment	Observed result	Decision recorded
Prompt voting	0.851 local, 0.783 public	Stop prompt churn; require new evidence
TF-IDF verifier	0.826 out of fold	Keep as a portable fallback, not a scorer
Raw QA retrieval	Relevant passage, wrong relation	Retrieval proposes; it never decides
v87 fact edits	**0.966 to 0.965 public**	Require source + predicate; revert

Admission rule: a change must add an evidence trace and preserve the regression suite.

[7] Rejected and scored experiment records in EXPERIMENT.log

Reproducibility was treated as an evaluation result

Correctness, deterministic execution, output integrity, and runtime were checked separately.

249
automated regression tests

3
byte-identical inference runs

225
deterministic controls

0
schema or ID-order failures

Literal path — organizer test path → **Pinned assets** — model + runtime + evidence → **Fixed schedule** — rows + prompts + batches → **T4×2 inference** — one tensor-split server → **Validated CSV** — ordered id,label

3 runs matched byte-for-byte • scored CSV hash matched • 5,000 Phase 2 rows validated in 3:52:02

[7] Regression suite, freeze manifest, evaluation ledger [8] Phase 2 runtime evidence

Phase 2 completed in 3 h 52 min with 95.9% judged

Measured organizer-run inference on 5,000 rows using Kaggle Tesla T4 x2.

5,000 held-out rows

207 deterministic (4.14%)

4,793 Qwen judge (95.86%)

3:52:02 validated end to end

Measured milestones	Elapsed	Observation
Router complete	1:13.18	207 resolved; 4,793 abstained
Model ready	2:51.25	16.332 GiB Qwen across two T4 GPUs
Judge complete	3:52:01.91	13,747.50 s; 0.349 rows/s

42.97% of the 9-hour budget used | 5:07:58 headroom

[8] Organizer-executed Phase 2 notebook and runtime report

Phase 2 source shift collapsed deterministic coverage

Exact Phase 1 lineage will cover less. The fallback must remain safe when coverage drops.

Web and news

Common Crawl, recent newspapers

Knowledge

Wikipedia and Banglapedia

Civic and legal

government services and bdlaws

Education and culture

NCTB textbooks and literature

Try exact routing

context clauses
structured solvers
entity-relation checks

If uncertain

skip lineage
use the fixed judge
keep the same vote rule

Deterministic coverage: 93.72% in Phase 1 -> 4.14% in Phase 2

The fixed judge absorbed the shift and still finished in 3:52:02.

What the experiments support—and what remains open

Supported

Task families need separate policies.
Predicate alignment prevents relation swaps.
Abstention protected runtime under shift.

Limitations

Only 299 released labels.
Aggregate splits provide no row-level errors.
Phase 2 accuracy is still unknown.

Next analyses

Inspect Phase 2 errors when released.
Calibrate the judged tail.
Expand licensed civic and news evidence.

Questions

Selective inference remained reproducible and completed the held-out run within the runtime limit.